

Data Card

- **Input vs. Output:** The model's input consists of short, natural language questions from farmers, and the output is a ranked CSV list containing the QueryId and the top-5 predicted DocumentIds.
- **Classification Task:** The dataset utilizes graded relevance rather than simple binary classification. Documents are scored as 3 (Perfect), 2 (Relevant), 1 (Marginal), or 0 (Not relevant/hard negatives).
- **Labelling Strategy:** Ground truth labels are provided in the qrels_train.csv file. The system's ranking quality is strictly evaluated using the nDCG@5 metric to ensure the highest quality advice appears first.
- **Data Creation & Provenance:** The 695-document knowledge base was provided by the Kaggle competition hosts. Our research showed the corpus originates from two sources: open-access CC-BY agricultural research (e.g., CGIAR) and synthetic public-domain documents.
- **Web Collection Ethics:** We checked and verified that the dataset adopted ethical sourcing. All included "grounded" texts maintain strict adherence to CC-BY licensing by excluding non-commercial or no-derivatives materials and retaining attribution links to the original authors.
- **Bias and Representation:** The dataset specifically captures the semantic differences between expert agronomic jargon and conversational farmer queries. This design choice mitigates the bias of traditional lexical systems that only serve users who know exact scientific terms.
- **Values and Community Alignment:** Aligning with our core value of Resource-Consciousness, the dataset allows for the training of lightweight embedding and retrieval models. This guarantees the resulting application can operate offline on standard CPU hardware, directly supporting agricultural resilience for under-resourced farming communities.